

Subject Description Form

Subject Code	COMP5565
Subject Title	Decentralized Apps Fundamentals and Development
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: 1. introduce the motivation of decentralized application (DApp) and how it can be realized by blockchain and smart contracts; 2. present a systematic approach to planning, designing, implementing and testing DApps in various use cases with proper blockchain network configuration; 3. introduce common blockchain frameworks and how they can be applied in appropriate settings.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: 1. understand the architecture and components of a DApp, including the front-end interface and back-end processing, supported by blockchain and smart contracts; 2. plan, design, implement and test end-to-end DApps with proper blockchain network configuration in enterprise scale; 3. critically review the newly emerging blockchain standards and architectures, and apply them in various use cases.
Subject Synopsis/ Indicative Syllabus	<u>Blockchain and DApps Fundamentals</u> 1. History of distributed computing 2. Consensus algorithms and protocols 3. Permission-less and permissioned blockchains 4. Case studies <u>Blockchain DApps Frameworks and Development</u> 5. Ethereum - Architecture - Wallet - Gas and gas price - Smart contract - Network configuration, maintenance and testing

	6. Hyperledger Fabric - Architecture - Chaincode basics and lifecycle - Identity management and Membership Service Provider (MSP) - Private data collection - Network configuration, maintenance and testing 7. Other emerging frameworks <u>Other Considerations in DApp Development</u> 8. Token standards 9. Hybrid chain 10. Security concerns																																				
Teaching/Learning Methodology	This course will emphasize on both the theories and practical aspects of DApp fundamentals and development. The theories will be covered through lectures, whereas the practice aspects will be taught through a series of laboratories which are designed to help student understand the processing of designing and implementing decentralized applications using various blockchain frameworks.																																				
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed</th></tr> <tr> <th>a</th><th>b</th><th>c</th></tr> </thead> <tbody> <tr> <td>1. Continuous Assessment</td><td></td><td></td><td></td><td></td></tr> <tr> <td>• Mid-term Test</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>• Project</td><td>15%</td><td></td><td>✓</td><td></td></tr> <tr> <td>2. Examination</td><td>70%</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Total</td><td>100%</td><td colspan="3"></td></tr> </tbody> </table> The mid-term test and examination are designed to evaluate the students' understanding on the theories of decentralized application. The project, on the other hand, are designed to evaluate the students' practical skills on developing DApps and setting up Blockchain networks.				Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			a	b	c	1. Continuous Assessment					• Mid-term Test	15%	✓	✓	✓	• Project	15%		✓		2. Examination	70%	✓	✓	✓	Total	100%			
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Student Study Effort Expected	Class contact:																																				
	▪ Lectures and Laboratory		39 Hrs.																																		
	Other student study effort:																																				
	▪ Self-study		66 Hrs.																																		

	Total student study effort	105 Hrs.
Reading List and References	Reference Books: 1. D. Cawrey, L. Lantz, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, O'Reilly Media 2020. 2. A. M. Antonopoulos, Mastering Ethereum: Building Smart Contracts and DApps, Mastering Ethereum: Building Smart Contracts and DApps, Sebastopol, CA: O'Reilly Media, Inc. 2019. 3. N. Gaur, A. O'Dowd, P. Novotny, L. Desrosiers, V. Ramakrishna, S. A. Baset, Blockchain with Hyperledger Fabric: Build decentralized applications using Hyperledger Fabric 2, 2nd ed., Packt Publishing 2020. 4. N. R. Thota, Mastering Hyperledger Fabric: Master The Art of Hyperledger Fabric on docker, docker swarm and Kubernetes, 2020.	